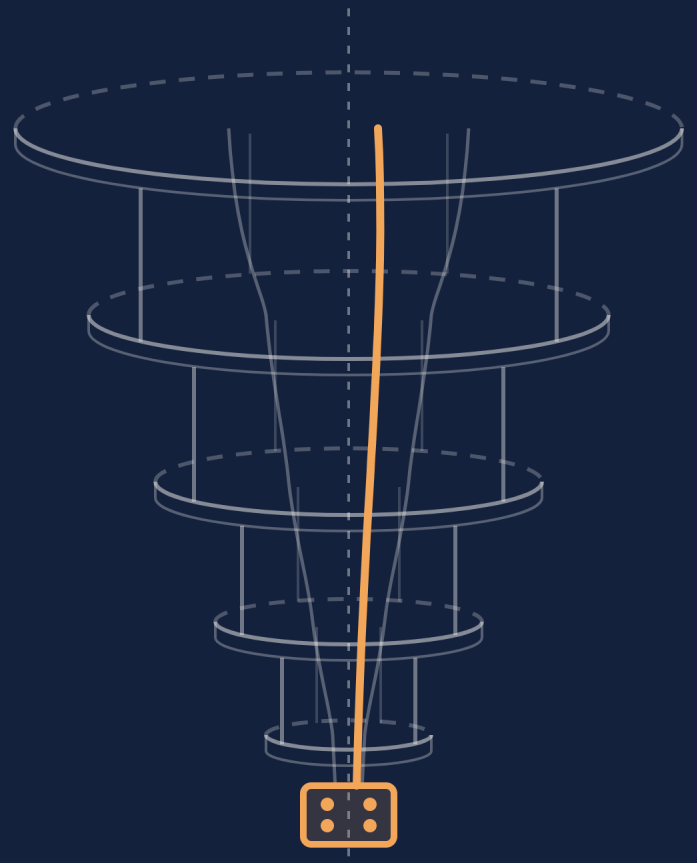
QUANTUM COMPUTING EXPLAINED

The Map of Quantum Computing

How quantum computers work, what they could do, and how people are building them



Louis Libat - October 2026

OVERVIEW

The map at a glance

How it works

Superposition · entanglement · interference

Models

Circuit · measurement-based · adiabatic · annealing · topological

Quantum computing

What it could do

Shor & Grover · complexity · quantum simulation · optimisation

Implementations

Superconducting · quantum dots · photonic · trapped ions · colour centres · neutral atoms

Obstacles · decoherence · noise · error correction · scalability

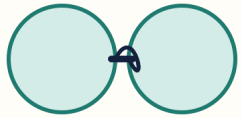
HOW IT WORKS

Three ingredients



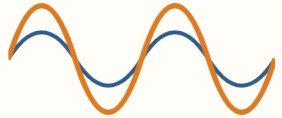
Superposition

A qubit can be in a combination of 0 and 1, not just one or the other.



Entanglement

Qubits join into one large shared state and are no longer independent.



Interference

Wavefunctions add up, so amplitudes can reinforce or cancel.

Classical computer: **one state at a time**

Quantum computer: **a huge number of states at once**

HOW IT WORKS

Bits versus qubits

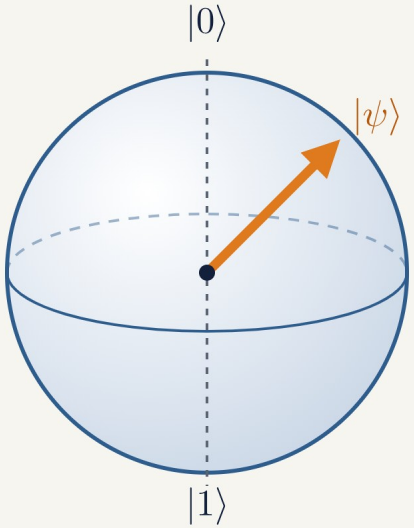
Classical bit: a switch

Either 0 or 1. Measuring it simply returns the state it is in.



Qubit: an arrow on a sphere

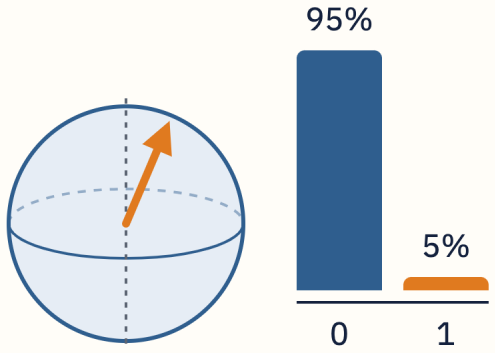
Up is 0 and down is 1, like a bit. Any other direction is a superposition.



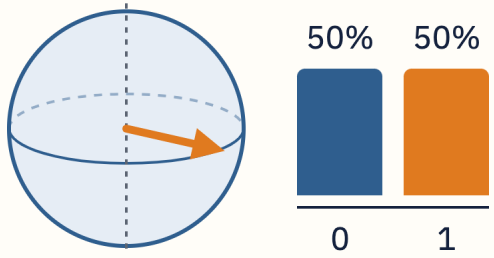
HOW IT WORKS

Superposition and measurement

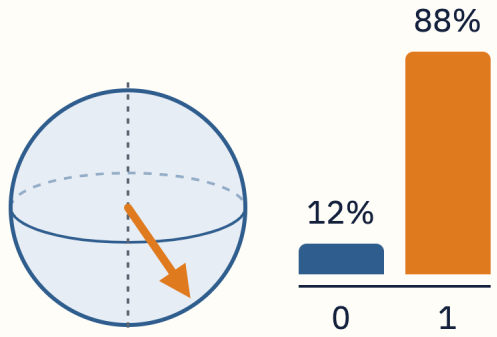
Measuring a qubit always gives **0 or 1**. Which one is random, with a probability set by the direction of the arrow.



Mostly up → mostly **0**



On the equator → **50 / 50**



Mostly down → mostly **1**

HOW IT WORKS

Entanglement: one large shared state

- Unentangled qubits have **independent** probabilities.
- Once entangled, we need one joint distribution over **every** outcome: 00, 01, 10, 11.
- Rotating one qubit changes the distribution of the whole system.



Example joint distribution of two entangled qubits

HOW IT WORKS

Every qubit doubles the state space

Qubits n	Basis states 2^n
1	2
2	4
3	8
10	$1\,024$
50	$\approx 1.1 \times 10^{15}$
300	$\approx 2 \times 10^{90}$

THE CORE DIFFERENCE

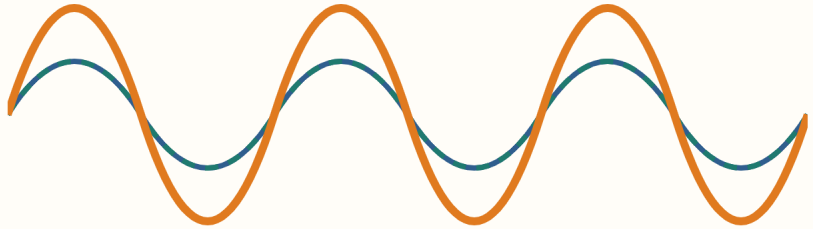
A classical register holds *any one* of these states.

n qubits can be in a superposition of all 2^n at once.

HOW IT WORKS

Interference: adding waves

The state of a quantum computer is a **wavefunction**. Entangled qubits' wavefunctions add into one, and adding waves means interference.



Constructive

Waves in step add up to a bigger wave.



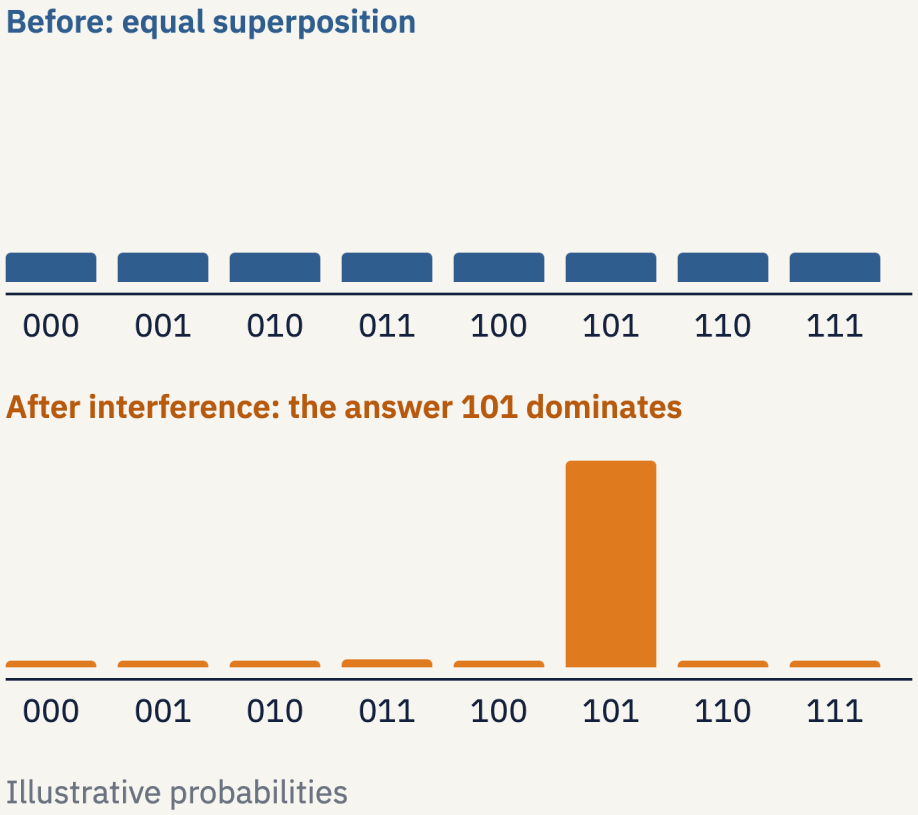
Destructive

Opposite waves cancel each other out.

HOW IT WORKS

Using interference to compute

- However many states are in the superposition, a measurement returns **just one**.
- An algorithm uses **constructive** interference on the right answer and **destructive** interference on the wrong ones.
- Then the right answer is what you most likely measure.

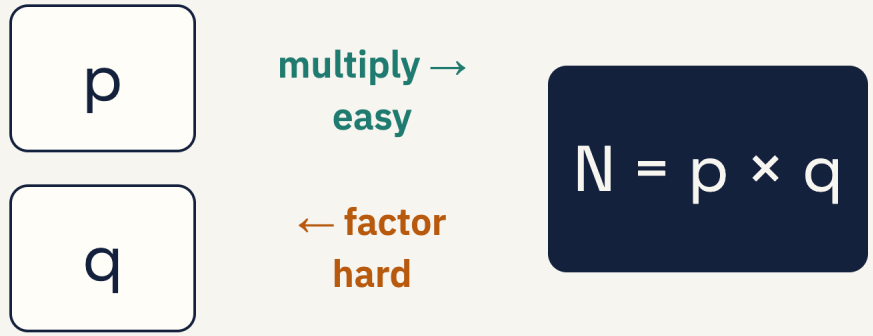


Illustrative probabilities

WHAT IT COULD DO

Shor’s algorithm (1994)

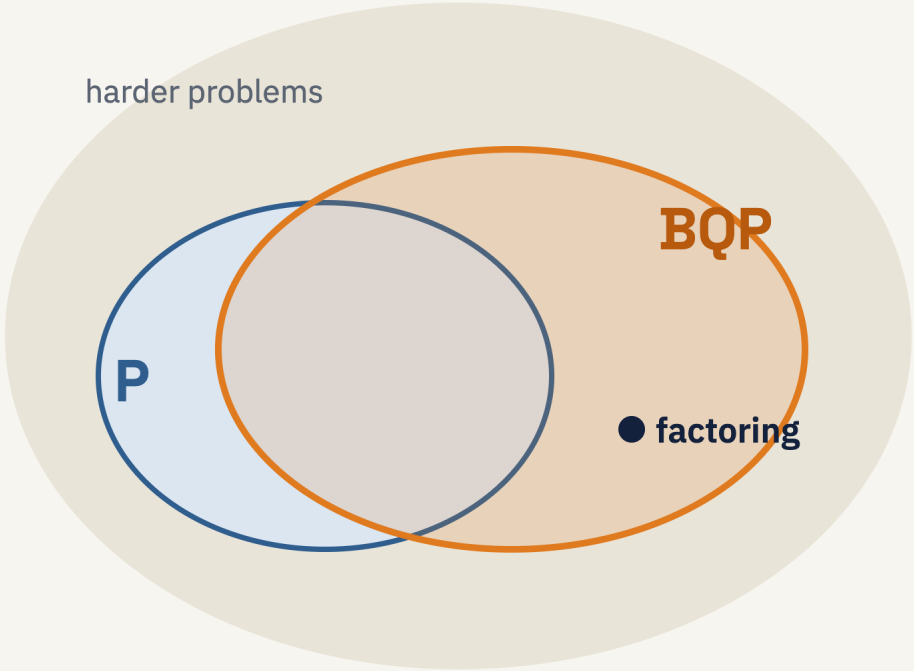
- Multiplying two large numbers is **easy**.
- Finding the **factors** again is hard: the search space is huge and no efficient classical algorithm is known.
- That one-way difficulty secures websites, email and banking.
- Peter Shor's efficient quantum algorithm was the first application with major real-world stakes.



For example 3,233 = 61 × 53, but going backwards takes a search.

WHAT IT COULD DO

Complexity: how hard does a problem get?

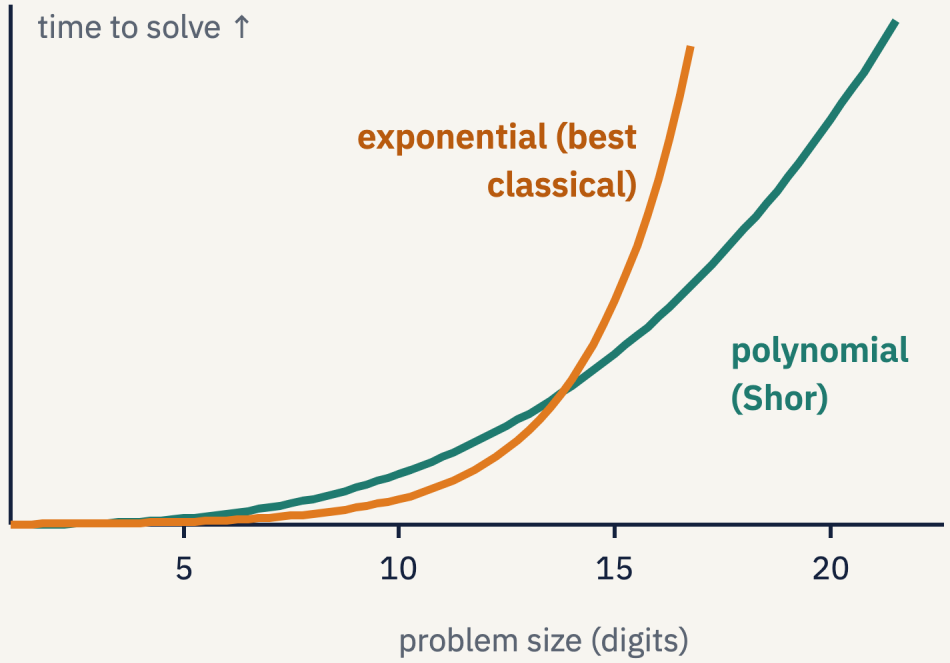


- Complexity theory sorts problems by how fast they get harder as the input grows.
- **P**: efficient on a classical computer.
- **BQP**: efficient on a quantum computer.
- BQP problems outside P are where quantum computers win, and factoring is believed to be one.

Schematic only: the exact relationships between these classes are not proven.

WHAT IT COULD DO

Exponential versus polynomial scaling



- Add one digit: how much harder does it get?
- With the size in the exponent, each extra digit **multiplies** the cost.
- Shor turns factoring from exponential into **polynomial**, so an intractable problem becomes solvable on a working quantum computer.

Illustrative curves, not real timings.

WHAT IT COULD DO

Your bank account is safe, for now

≈100

qubits in the most advanced universal machines

≈1,000,000

qubits estimated to run Shor on real encryption keys

Post-quantum encryption

Schemes that do not rely on factoring

Quantum cryptography

Quantum physics itself secures the channel

Other algorithms

Different speed-ups, e.g. Grover's search of unstructured lists

Qubit numbers as estimated in the video (2021).

WHAT IT COULD DO

Don’t underestimate classical computers

Same computing power, different algorithms

- A clever classical factoring algorithm is thought unlikely, but not ruled out.
- **Non-computable** problems, like the halting problem, defeat both kinds of computer.
- Each can simulate the other.

But simulation is lopsided

Simulating a quantum computer on a classical one gets **exponentially harder** with every qubit.

A quantum computer *is* a quantum system, so it does not have that problem.

WHAT IT COULD DO

Quantum simulation

~30

particles are already enough to strain the largest supercomputers.

The goal: prototype materials **inside a quantum computer** instead of building and testing each one in a lab.

Superconductors
exotic materials at low temperature

Fertiliser
its production emits ~2% of global CO_2

Batteries
better chemistries

Solar panels
more efficient materials

Drugs & chemicals
new molecules and reactions

Aerospace
new structural materials

WHAT IT COULD DO

Other applications, and a word on hype

Proposed applications

- Optimisation problems
- Machine learning and AI
- Financial modelling
- Weather forecasting and climate
- Cybersecurity, mostly via Shor

Keep it in perspective

- These are all *potential* uses: no quantum computer yet beats classical ones on a real-world problem.
- Many claims come from people raising money to build these machines.
- Early computer pioneers never imagined the internet, so the best uses may still be unknown.

MODELS OF QUANTUM COMPUTING

Two layers of nuance

Models

The overall approach to manipulating a collection of qubits: circuit, measurement-based, adiabatic, annealing, topological



Physical implementations

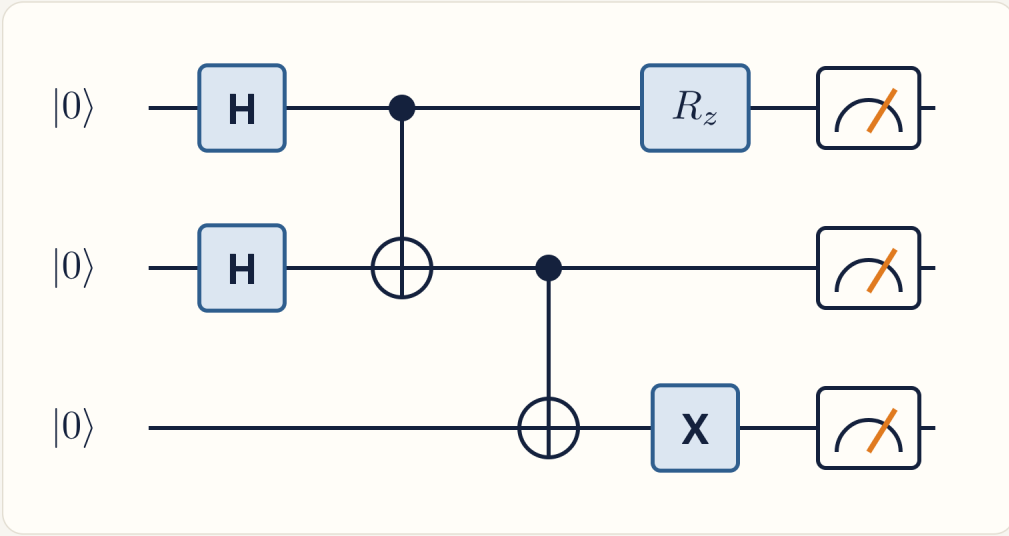
The actual quantum objects the qubits are built from: superconducting loops, electrons in silicon, photons, ions, atoms

Classical computers have practically one model, bits plus logic gates. Quantum computing has several.

MODELS

The circuit (gate) model

- Classical analogue: bits plus logic gates. NOT and NOR alone can build a whole computer.
- Quantum **gates** act on a few qubits, rotating their arrows without measuring them.
- An algorithm is a sequence of gates, then a **measurement**.
- The most popular and best understood model.



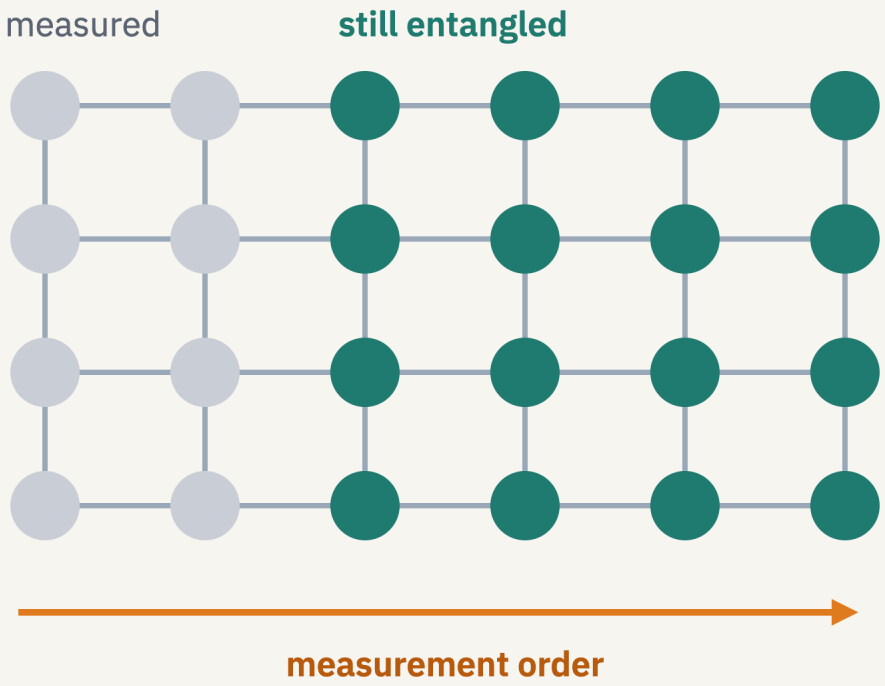
Qubits on wires, gates in boxes, time flows left to right

Hands-on: the video recommends IBM's free, open-source Qiskit textbook and SDK (the video's sponsor).

MODELS

Measurement-based (one-way) computing

- First prepare a large **entangled resource state**.
- Then compute by measuring the qubits **one by one**.
- Mathematically equivalent to the circuit model.



MODELS

Adiabatic quantum computing

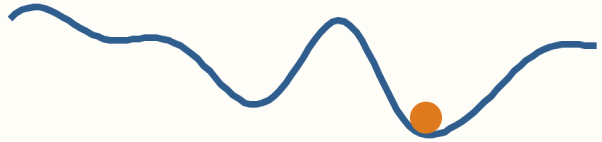
Physical systems always move towards their **minimum energy**. So pose the problem so that its **lowest-energy state is the answer**.



1. Start flat



2. Slowly bring in the problem



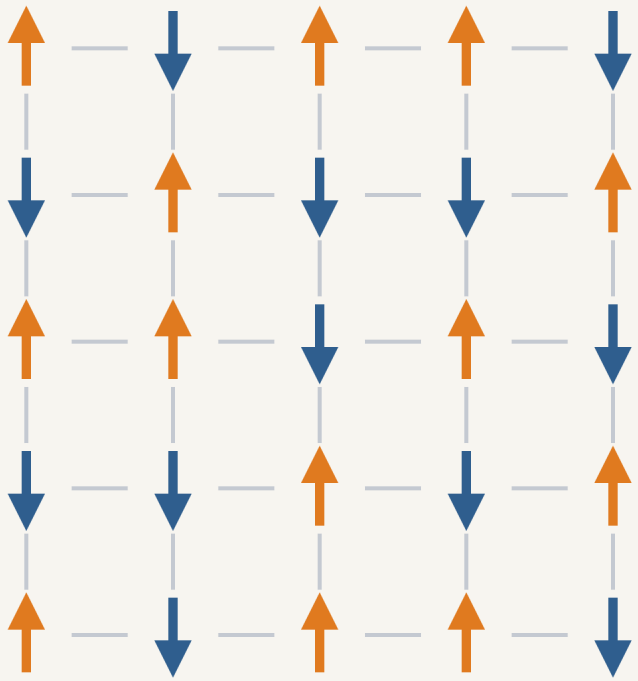
3. Lowest point = the answer

Go slowly enough and the system stays in its lowest energy state. The model is **universal** and equivalent to the circuit model.

MODELS

Quantum annealing

- Same principle as adiabatic computing: find the minimum of an energy landscape.
- **Not universal**: it lacks the freedom to represent every quantum state.
- Still useful for **optimisation** and for simulating some quantum systems, such as spin glasses.
- A stepping stone towards a universal adiabatic computer.



Spin glass: coupled magnetic spins (orange up, blue down)

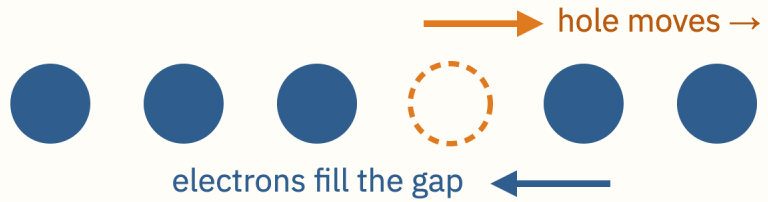
The video's author previously worked at D-Wave, which builds quantum annealers.

MODELS

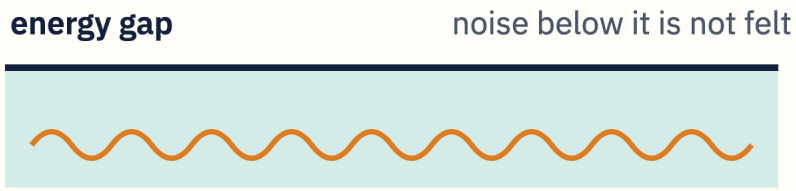
Topological quantum computing

- The most theoretical model: qubits from **Majorana zero modes**, a kind of non-abelian anyon.
- **Quasi-particles** emerge from the collective behaviour of many particles.
- Whether Majoranas have been observed is still debated.
- Expected to be very stable: separated parts plus an **energy gap** that blocks low-energy noise.

The electron hole: a quasi-particle



Protected by an energy gap



BUILDING REAL MACHINES

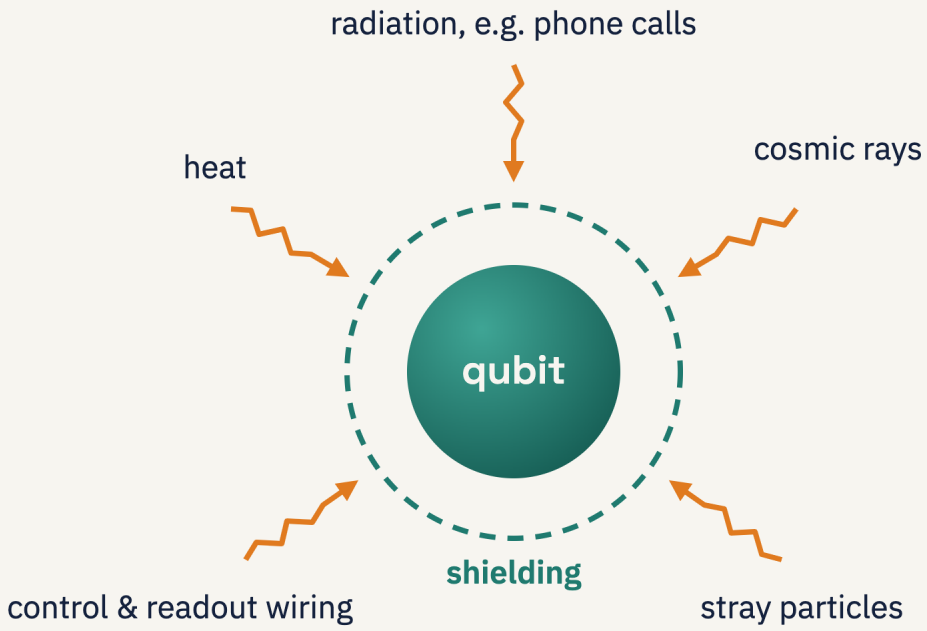
What it takes to make a qubit

The requirement

Any **two-state quantum system**: one state is 0, the other 1. The obvious example is a particle's spin, up or down.

The catch: decoherence

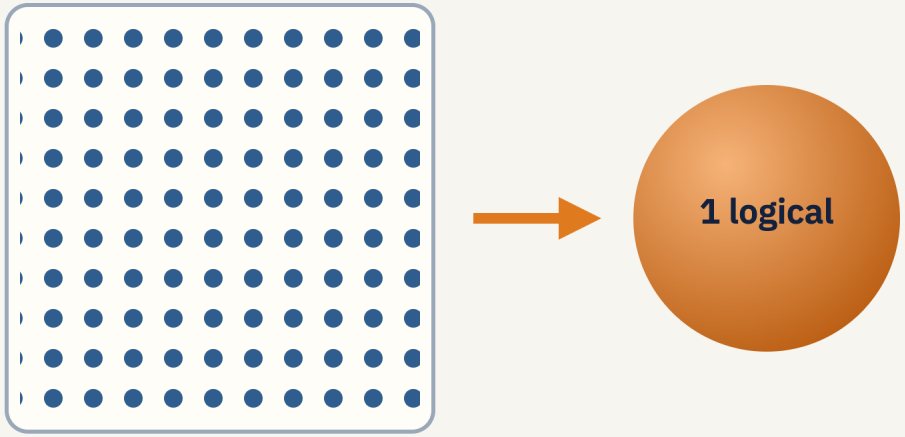
Any contact with the outside world leaks the information away. Qubits are “dumb” and entangle with anything they can.



BUILDING REAL MACHINES

Error correction and scalability

- Some noise is **inevitable**, and it grows with the number of entangled qubits.
- **Quantum error correction**: many physical qubits act as one noise-free logical qubit.
- **Scalability**: every qubit needs its own control and readout wiring, a massive engineering problem.



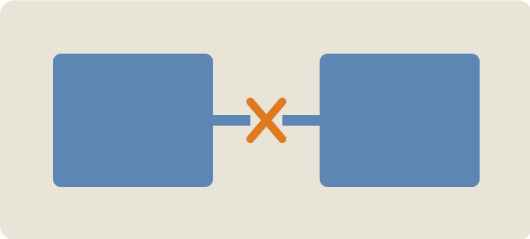
~100 physical qubits

100 – 1000 physical per logical qubit

The open question for the field: can a large, working quantum computer beat decoherence and noise?

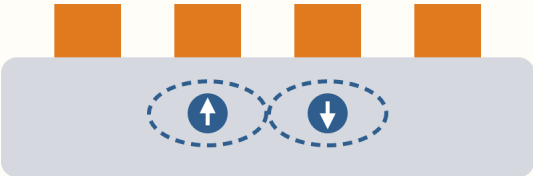
BUILDING REAL MACHINES

Platforms: circuits, electrons, light



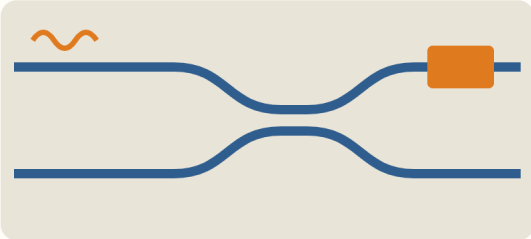
Superconducting

- Currently the most popular approach
- A superconducting circuit broken by a Josephson junction
- Transmon: charge oscillating across the junction; also flux and phase qubits



Quantum dots / silicon spin

- Built from semiconductors such as silicon
- An electron's spin or charge, trapped in a tiny quantum dot
- Controlled with voltages, microwaves or magnetic fields

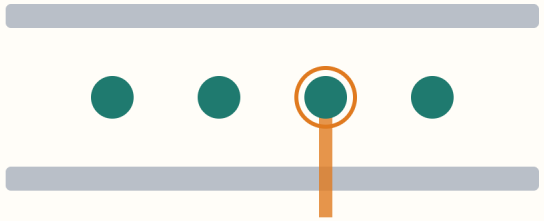


Photonic

- Qubits are photons in integrated photonic chips
- Encoded in the paths a photon takes, or photon numbers
- Mirrors, waveplates and interferometers; voltages on paths

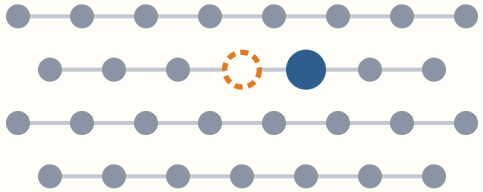
BUILDING REAL MACHINES

Platforms: atoms and ions



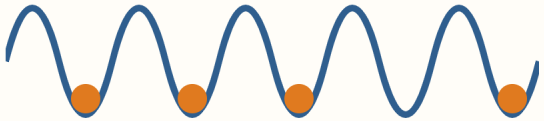
Trapped ions

- Charged atoms, levitated and moved by electromagnetic fields
- Qubit: two specific energy levels of the atom
- Controlled and read out with microwaves or lasers



Colour centres

- Atoms embedded in a solid, e.g. nitrogen (blue) beside a vacancy (dashed) in diamond
- Qubit: nuclear spins of the embedded atoms
- Entangled together via electrons



Neutral atoms

- Atoms like caesium in an “egg box” of laser beams, cooled to millionths of a kelvin
- Qubit: hyperfine or Rydberg levels, controlled by lasers
- Also simulators: one with 10,000 atoms has been built

BUILDING REAL MACHINES

Six platforms, side by side

Platform	Qubit (two-level system)	Control
Superconducting	charge, flux or phase at a Josephson junction	microwaves
Quantum dots	spin or charge of electrons in silicon	voltages, microwaves
Photonic	paths or photon numbers of light	optics, voltages
Trapped ions	two energy levels of a levitated ion	microwaves, lasers
Colour centres	nuclear spin of an embedded atom	via electrons
Neutral atoms	hyperfine or Rydberg levels of cold atoms	lasers

Also explored, at smaller scale: electrons on helium, cavity QED, magnetic molecules, molecular spins, NMR.

SUMMARY

Takeaways

1. **Superposition, entanglement, interference**: n qubits span 2^n states, and algorithms steer interference towards the right answer.

2. The promise is in **specific problems** (factoring, search and above all quantum simulation), not in being faster at everything.

3. Several **models** (circuit, measurement-based, adiabatic, annealing, topological) run on many **physical platforms**.

4. Decoherence, noise, error correction and scaling are the obstacles, and it is **not yet clear which approach will win**.

The race to build a useful quantum computer is still wide open.

Source: CERFACS - BULL Training - October 2026